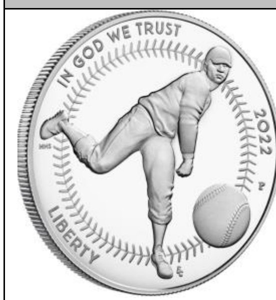
Grade 8
Unit 14 Assessment
Calculator

Name____________________
Date_________________
Period__________

1. In 2022, the Negro Leagues Baseball Silver Dollar was released by the U.S. Mint as part of the Negro Leagues Baseball Commemorative Coin Program. The faces of the coin are shown.

2022 Negro Leagues Commemorative Silver Dollar	
Pitcher	**Batter**
	

a. Determine the sample space. Represent the sample space using an organized list, a table, or a tree diagram.

Repeated Experiment	**Sample Space**
Flipping the Negro Leagues Silver Dollar 3 times.	

b. Determine the theoretical probability of getting pitcher, batter, batter. Write your answer as a percentage rounded to the nearest tenth of a percent.

$\boldsymbol{P}(\textbf{pitcher}, \textbf{batter}, \textbf{batter})$	

2. A fair $17-$ sided number cube is shown.

a. How many outcomes are in the sample space when rolling the number cube twice?

work	
Number of Outcomes	

b. Determine the theoretical probability of rolling a number that is divisible by 4 two times in a row. Write the answer as a decimal rounded to the nearest thousandth.

work	
$\boldsymbol{P(}$ **number that is divisible by** $\boldsymbol{4,}$ **number that is divisible by** $\boldsymbol{4)}$	

3. A bag contains 2 purple marbles, 1 green marble, and 3 yellow marbles. A marble is randomly drawn from the bag twice, with replacement. Determine the theoretical probability of choosing a green marble on both draws. Write your answer as a fraction.

work	
$\boldsymbol{P}(\mathbf{green}, \mathbf{green})$	

4. Each letter in the word **KISSIMMEE** is written on a card and put into a bag. Determine the theoretical probability of choosing an "**S**", putting it back and then drawing and "**I**". Write your answer as a fraction.

work	
$\boldsymbol{P}(\mathbf{S}, \mathbf{I})$	

5. Syndney has 4 watermelon jellybeans, 3 sour apple jellybeans, 2 tutti-frutti jellybeans, and 3 cotton candy jellybeans. As an experiment, she puts the jellybeans into a bag and randomly draws from the bag twice, with replacement. She does this 144 times. Her results are shown.

Outcome	Number of Times	Outcome	Number of Times
Cotton Candy, Cotton Candy	9	Tutti-Frutti, Cotton Candy	6
Cotton Candy, Sour Apple	9	Tutti-Frutti, Sour Apple	8
Cotton Candy, Tutti-Frutti	8	Tutti-Frutti, Tutti-Frutti	3
Cotton Candy, Watermelon	12	Tutti-Frutti, Watermelon	6
Sour Apple, Cotton Candy	9	Watermelon, Cotton Candy	12
Sour Apple, Sour Apple	12	Watermelon, Sour Apple	9
Sour Apple, Tutti-Frutti	8	Watermelon, Tutti-Frutti	7
Sour Apple, Watermelon	10	Watermelon, Watermelon	16

Using the results from the experiment, determine the experimental probability of each outcome. Write your answer as a decimal rounded to the nearest thousandth. Show your work.

	Outcome	work	Experimental Probability
a.	Sour Apple, Watermelon		$P(\text{Sour Apple, Watermelon}) =$ ____________

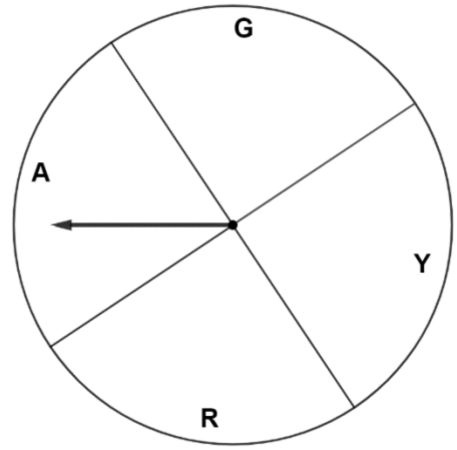
b.	Watermelon, Cotton Candy		$P(\text{Watermelon, Cotton Candy}) =$ ______________

c. Select all of the experimental probablities of Syndney’s experiment that also match the theoretical probability.

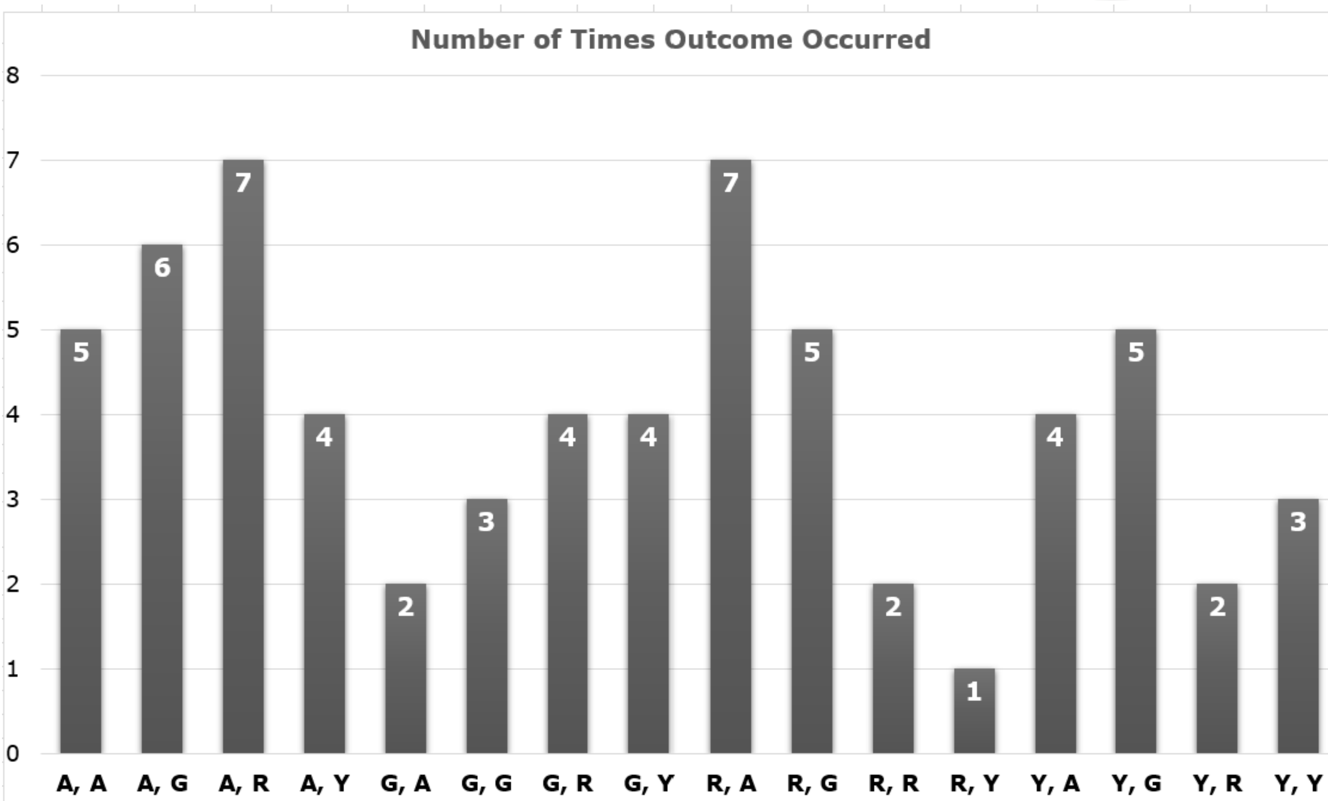
- ☐ $P(\text{Cotton Candy, Cotton Candy})$
- ☐ $P(\text{Sour Apple, Tutti} - \text{Frutti})$
- ☐ $P(\text{Sour Apple, Watermelon})$
- ☐ $P(\text{Tutti} - \text{Frutti, Cotton Candy})$
- ☐ $P(\text{Watermelon, Cotton Candy})$

work

6. A fair spinner is shown.



In an experiment, Elizabeth spun the spinner twice 64 times and recorded the outcomes in a bar chart as shown.



a. Determine the experimental probability of $P(A,G)$. Write your answer as a percentage rounded to the nearest tenth of a percent.

work	
$\boldsymbol{P(A,G)}$	

b. Determine which outcome(s) in Elizabeth's experiment have the same experimental probability as its theoretical probability.

work	
Outcome(s)	

c. Choose the best phrase to complete each statement.

If, in a second experiment, Elizabeth spun the spinner twice 32 times, she would expect the new results to

- have experimental probabilities that are different than the first experiment.
- have exactly the same experimental probabilities as in the first experiment.
- have experimental probabilities that are closer to the theoretical probabilities.
- have outcomes that are exactly $\frac{1}{2}$ of the outcomes in the first experiment.

If, in a second experiment, Elizabeth spun the spinner twice 640 times, she would expect the new results to

- have exactly the same experimental probabilities as in the first experiment.
- have experimental probabilities that are closer to the theoretical probabilities than the first experiment.
- have outcomes that are exactly 10 times the outcomes in her first experiment.
- have experimental probabilities that are exactly twice as much as in the first experiment.